

# Advay Bhagat

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## EDUCATION

**Vellore Institute Of Technology**  
*Bachelor of Technology in Computer Science*

CGPA: 8.45  
*Expected May 2028*

## EXPERIENCE

### Python Summer Intern

May 2026 – Present

*IIT Bombay*

*Mumbai, MH*

- Engineered backend architecture for the Yaksh online testing platform using Django and Python, optimizing RESTful APIs to support high-concurrency exam sessions and deliver a massively scalable infrastructure.
- Proactively guarded against cyber threats and academic malpractice by architecting a Safe Exam Browser (SEB) integration, implementing cryptographic hash validation to enforce strict security protocols and risk management.
- Designed innovative, low-latency solutions for real-time exam evaluation by deploying an asynchronous, containerized grading pipeline utilizing Docker and Celery to securely execute untrusted code.
- Collaborated within a fast-paced, Agile open-source environment, driving continuous integration and rigorous code reviews to rapidly adapt and deploy robust software solutions.

## PROJECTS

### KiranaIQ | *Python, Flask, React, PostgreSQL, Docker* | [GitHub](#) | [Live](#)

March 2026 – Present

- Built a full-stack retail analytics platform with a Flask REST API backend and React frontend, enabling kirana store owners to track inventory levels, sales trends, and stock movement in real time
- Designed a PostgreSQL schema for multi-product inventory management; integrated Celery with Redis for asynchronous restocking alerts and scheduled sales report generation
- Built a data analytics module processing historical sales records to generate demand forecasts and low-stock alerts, reducing manual inventory checks
- Implemented JWT-based authentication and role-based access control to support multiple store owner accounts with isolated data views and secure API endpoints

### MacOS Web Portfolio | *React, Tailwind CSS, Zustand, GSAP, Vite*

February 2025 – Present

- Built an interactive web portfolio simulating the macOS desktop experience with a custom window management system supporting drag-and-drop, minimize, and maximize
- Managed global application state with Zustand across open windows, active apps, and user preferences; integrated GSAP animations and Tailwind CSS for pixel-perfect responsive styling

### AI Sports Analytics Platform | *Python, SciPy, Monte Carlo, NLP, Django, React*

June 2026

- Engineered a full-stack FIFA World Cup prediction platform using Django and Next.js, processing datasets for 48 nations including ELO ratings, squad market values, and historical form
- Built a custom ML engine using Poisson distributions and Monte Carlo simulations (10,000 iterations/run) to forecast match outcomes and tournament winner probabilities
- Architected a real-time NLP sentiment pipeline scraping live football news, scoring sentiment from -1.0 to +1.0, and dynamically adjusting team xG by up to  $\pm 5\%$

## TECHNICAL SKILLS

**Languages:** Python, JavaScript/TypeScript, SQL (PostgreSQL), Java, C/C++, HTML/CSS

**Frameworks:** React, Next.js, Django, Flask, Node.js, Tailwind CSS, FastAPI

**Developer Tools:** Git, Docker, Redis, Celery, Vite, Vercel, Google Cloud Platform (GCP), VS Code

**Libraries & ML:** SciPy, Pandas, NumPy, Zustand, GSAP, Framer Motion, Recharts

**Systems & CS Fundamentals:** OS, DBMS, Data Structures and Algorithm